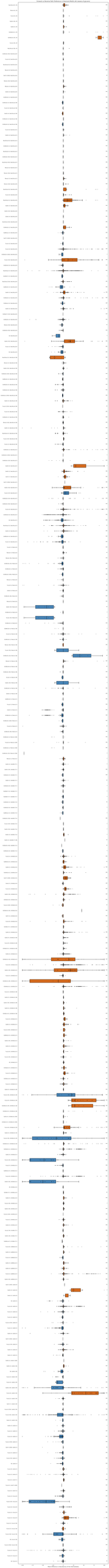

Forward vs Reverse Path Preference in Diamond Motifs (All classes of glycans)
Neu5Ac(a2-6), OS
Man(a1-6), OS
Fuc(a1-6), OS
Gal(b1-3/4), OS
Gal(b1-4), OS
GlcNAc(b1-2), OS
GlcNAc(b1-2/4), OS
Fuc(a1-3/4), OS
Neu5Ac(a2-3/6), OS
GlcNAc(b1-4/6), Neu5Gc(a2-6)
Fuc(a1-6), Neu5Gc(a2-6)
Neu5Ac(a2-6), Neu5Gc(a2-6)
Man(a1-6), Neu5Gc(a2-6)
Gal(?1-3/4/6), Neu5Gc(a2-6)
Man(a1-3/6), Neu5Gc(a2-6)
Man(a1-3), Neu5Gc(a2-6)
Gal(b1-4), Neu5Gc(a2-6)
GlcNAc(b1-2), Neu5Gc(a2-6)
GlcNAc(b1-4), Neu5Gc(a2-3/6)
Fuc(a1-6), Neu5Gc(a2-3/6)
Gal(b1-4), Neu5Gc(a2-3/6)
GlcNAc(b1-2), Neu5Gc(a2-3/6)
GlcNAc(b1-6), Neu5Gc(a2-3/6)
Fuc(a1-6), Neu5Gc(a2-3)
Gal(b1-4), Neu5Gc(a2-3)
Neu5Ac(a2-6), Neu5Gc(a2-3)
GlcNAc(b1-2), Neu5Gc(a2-3)
Neu5Ac(a2-3), Neu5Gc(a2-3)
GlcNAc(b1-4/6), Neu5Gc(a2-3)
Neu5Ac(a2-3/6), Neu5Gc(a2-3)
Man(a1-6), Neu5Gc(a2-3)
Man(a1-3/6), Neu5Gc(a2-3)
Man(a1-3), Neu5Gc(a2-3)
Man(a1-3/6), Neu5Ac(a2-8)
Neu5Ac(a2-6), Neu5Ac(a2-8)
Fuc(a1-6), Neu5Ac(a2-8)
Neu5Ac(a2-3), Neu5Ac(a2-8)
Gal(b1-3), Neu5Ac(a2-8)
Gal(b1-3/4), Neu5Ac(a2-8)
Gal(b1-4), Neu5Ac(a2-8)
Neu5Ac(a2-3/6), Neu5Ac(a2-8)
GlcNAc(b1-2), Neu5Ac(a2-8)
GalNAc(a1-3), Neu5Ac(a2-6)
Gal(?1-3/4/6), Neu5Ac(a2-6)
Fuc(a1-4), Neu5Ac(a2-6)
Fuc(a1-6), Neu5Ac(a2-6)
GlcNAc(b1-2/4/6), Neu5Ac(a2-6)
Fuc(a1-3/4), Neu5Ac(a2-6)
GlcNAc(b1-6), Neu5Ac(a2-6)
Neu5Ac(a2-3), Neu5Ac(a2-6)
HexNAc(?1-3), Neu5Ac(a2-6)
Neu5Ac(a2-3/6), Neu5Ac(a2-6)
GalNAc(b1-4), Neu5Ac(a2-6)
Gal(b1-3), Neu5Ac(a2-6)
GlcNAc(b1-2), Neu5Ac(a2-6)
GlcNAc(a1-4), Neu5Ac(a2-6)
Gal(b1-4), Neu5Ac(a2-6)
GlcNAc(?1-3/4), Neu5Ac(a2-6)
GlcNAc(b1-4), Neu5Ac(a2-6)
Man(a1-3/6), Neu5Ac(a2-6)
GlcNAc(b1-4/6), Neu5Ac(a2-6)
3S, Neu5Ac(a2-6)
Gal(b1-3/4), Neu5Ac(a2-6)
Fuc(a1-3), Neu5Ac(a2-6)
6S, Neu5Ac(a2-6)
Neu5Ac(a2-3), Neu5Ac(a2-3/8)
Man(a1-3), Neu5Ac(a2-3/6)
Gal(b1-3/4), Neu5Ac(a2-3/6)
GlcNAc(b1-4/6), Neu5Ac(a2-3/6)
GalNAc(b1-4), Neu5Ac(a2-3/6)
GlcNAc(b1-4), Neu5Ac(a2-3/6)
GlcNAc(b1-6), Neu5Ac(a2-3/6)
GlcNAc(b1-2/4/6), Neu5Ac(a2-3/6)
Gal(b1-4), Neu5Ac(a2-3/6)
Fuc(a1-2/3/4), Neu5Ac(a2-3/6)
Fuc(a1-6), Neu5Ac(a2-3/6)
GlcNAc(b1-2), Neu5Ac(a2-3/6)
Man(a1-3/6), Neu5Ac(a2-3/6)
Gal(b1-3), Neu5Ac(a2-3/6)
Neu5Ac(a2-3), Neu5Ac(a2-3/6)
Fuc(a1-3/4), Neu5Ac(a2-3/6)
Fuc(a1-3), Neu5Ac(a2-3/6)
GlcNAc(b1-2), Neu5Ac(a2-3)
GlcNAc(b1-2/4/6), Neu5Ac(a2-3)
GlcNAc(b1-2/6), Neu5Ac(a2-3)
Gal(b1-6), Neu5Ac(a2-3)
Gal(b1-3), Neu5Ac(a2-3)
Gal(?1-?), Neu5Ac(a2-3)
Gal(?1-3/4/6), Neu5Ac(a2-3)
Gal(b1-3/4), Neu5Ac(a2-3)
Fuc(a1-2/3), Neu5Ac(a2-3)
GlcNAc(b1-2/4), Neu5Ac(a2-3)
GlcNAc(b1-4/6), Neu5Ac(a2-3)
Fuc(a1-6), Neu5Ac(a2-3)
GlcNAc(b1-6), Neu5Ac(a2-3)
6S, Neu5Ac(a2-3)
Neu5Ac(a2-3), Neu5Ac(a2-3)
Gal(b1-4), Neu5Ac(a2-3)
GalNAc(b1-4), Neu5Ac(a2-3)
Fuc(a1-3), Neu5Ac(a2-3)
Fuc(a1-3), Man(a1-6)
Man(a1-3), Man(a1-6)
Man(a1-3/6), Man(a1-6)
GlcNAc(b1-2/4), Man(a1-6)
GlcNAc(b1-2), Man(a1-6)
GlcNAc(b1-2/4/6), Man(a1-6)
Man(a1-2), Man(a1-6)
Fuc(a1-6), Man(a1-6)
Gal(b1-4), Man(a1-6)
GlcNAc(b1-4/6), Man(a1-6)
Man(a1-6), Man(a1-6)
Gal(b1-3/4), Man(a1-6)
GlcNAc(b1-6), Man(a1-6)
Fuc(a1-3/4), Man(a1-6)
GlcNAc(b1-4), Man(a1-6)
GalNAc(b1-4), Man(a1-3/6)
Fuc(a1-6), Man(a1-3/6)
GlcNAc(b1-2), Man(a1-3/6)
Man(a1-3/6), Man(a1-3/6)
Fuc(a1-3/4), Man(a1-3/6)
GlcNAc(b1-2/4), Man(a1-3/6)
Man(a1-3), Man(a1-3/6)
GlcNAc(b1-6), Man(a1-3/6)
GlcNAc(b1-4/6), Man(a1-3/6)
Fuc(a1-3), Man(a1-3/6)
Gal(b1-3/4), Man(a1-3/6)
GlcNAc(b1-4), Man(a1-3/6)
Gal(b1-4), Man(a1-3/6)
GalNAc(b1-4), Man(a1-3)
Fuc(a1-3), Man(a1-3)
Gal(b1-4), Man(a1-3)
GlcNAc(b1-4), Man(a1-3)
GlcNAc(b1-2/4/6), Man(a1-3)
Fuc(a1-6), Man(a1-3)
GlcNAc(b1-4/6), Man(a1-3)
GlcNAc(b1-4), Man(a1-2/3/6)
Fuc(a1-6), Man(a1-2/3/6)
GlcNAc(b1-2), Man(a1-2/3/6)
GlcNAc(b1-2/4), Man(a1-2/3/6)
Man(a1-2), Man(a1-2)
Gal(b1-4), HexNAc(?1-?)
Gal(b1-3/4), HexNAc(?1-?)
GlcNAc(b1-3), HexNAc(?1-?)
Gal(b1-3), HexNAc(?1-?)
GlcNAc(b1-6), HexNAc(?1-?)
HexNAc(?1-4), HexNAc(?1-?)
HexNAc(?1-?), HexNAc(?1-?)
GlcNAc(b1-3), HexNAc(?1-4)
GlcNAc(b1-6), HexNAc(?1-4)
GalNAc(a1-3), HexNAc(?1-4)
GlcNAc(b1-3/6), HexNAc(?1-4)
Fuc(a1-3/4), HexNAc(?1-4)
Gal(b1-3/4), HexNAc(?1-4)
Gal(b1-3), HexNAc(?1-4)
Gal(b1-3), HexNAc(?1-3/6)
GlcNAc(b1-3/6), HexNAc(?1-3)
GlcNAc(b1-6), HexNAc(?1-3)
Fuc(a1-2), GlcNAc(b1-6)
GalNAc(b1-4), GlcNAc(b1-6)
Gal(b1-3), GlcNAc(b1-6)
GlcNAc(b1-2), GlcNAc(b1-6)
Gal(?1-3/4/6), GlcNAc(b1-6)
Gal(a1-3), GlcNAc(b1-6)
Fuc(a1-4), GlcNAc(b1-6)
Gal(b1-3/4), GlcNAc(b1-6)
GlcNAc(b1-6), GlcNAc(b1-6)
Fuc(a1-2/3/4), GlcNAc(b1-6)
GlcNAc(b1-2/4), GlcNAc(b1-6)
Gal(?1-3), GlcNAc(b1-6)
Fuc(a1-6), GlcNAc(b1-6)
GlcNAc(?1-?), GlcNAc(b1-6)
Gal(b1-4), GlcNAc(b1-6)
Gal(b1-4), GlcNAc(b1-4/6)
Fuc(a1-6), GlcNAc(b1-4/6)
GlcNAc(b1-2), GlcNAc(b1-4/6)
Gal(b1-4), GlcNAc(b1-4)
GlcNAc(b1-2/4), GlcNAc(b1-4)
Fuc(a1-6), GlcNAc(b1-4)
Gal(b1-3/4), GlcNAc(b1-4)
Fuc(a1-3), GlcNAc(b1-4)
GlcNAc(b1-2/6), GlcNAc(b1-4)
GlcNAc(b1-2), GlcNAc(b1-4)
Fuc(a1-2), GlcNAc(b1-3/6)
Gal(b1-3), GlcNAc(b1-3/6)
Gal(b1-3/4), GlcNAc(b1-3/6)
Gal(b1-4), GlcNAc(b1-3/6)
GlcNAc(a1-4), GlcNAc(b1-3/6)
Fuc(a1-4), GlcNAc(b1-3)
Gal(b1-3), GlcNAc(b1-3)
Fuc(a1-2/3/4), GlcNAc(b1-3)
GalNAc(b1-4), GlcNAc(b1-3)
Gal(b1-3/4), GlcNAc(b1-3)
Gal(b1-4), GlcNAc(b1-3)
Fuc(a1-6), GlcNAc(b1-3)
Fuc(a1-3/4), GlcNAc(b1-3)
6S, GlcNAc(b1-3)
GlcNAc(b1-3), GlcNAc(b1-3)
Fuc(a1-3), GlcNAc(b1-3)
GlcNAc(a1-4), GlcNAc(b1-3)
Fuc(a1-2), GlcNAc(b1-3)
Gal(b1-4), GlcNAc(b1-2/6)
Fuc(a1-3/4), GlcNAc(b1-2/6)
Fuc(a1-6), GlcNAc(b1-2/6)
GlcNAc(b1-2/4/6), GlcNAc(b1-2/6)
GlcNAc(b1-2), GlcNAc(b1-2/6)
Gal(b1-4), GlcNAc(b1-2/4/6)
GlcNAc(b1-2), GlcNAc(b1-2/4/6)
Fuc(a1-6), GlcNAc(b1-2/4)
Gal(b1-4), GlcNAc(b1-2/4)
Fuc(a1-3/4), GlcNAc(b1-2/4)
GlcNAc(b1-2), GlcNAc(b1-2/4)
GlcNAc(b1-2), GlcNAc(b1-2)
Gal(b1-4), GlcNAc(b1-2)
Fuc(a1-3/4), GlcNAc(b1-2)
Fuc(a1-3), GlcNAc(b1-2)
GalNAc(b1-4), GlcNAc(b1-2)
Fuc(a1-6), GlcNAc(b1-2)
Gal(b1-3/4), GlcNAc(b1-2)
6S, GlcNAc(a1-4)
GlcNAc(?1-?), GlcNAc(a1-4)
Gal(b1-3), GlcNAc(a1-4)
Fuc(a1-3/4), GlcNAc(a1-4)
Gal(b1-3/4), GlcNAc(a1-4)
Fuc(a1-2/3/4), GlcNAc(a1-4)
Fuc(a1-2), GlcNAc(a1-4)
Gal(b1-4), GlcNAc(a1-4)
Fuc(a1-3/4), GlcNAc(?1-?)
Gal(b1-3/4), GlcNAc(?1-?)
GalNAc(a1-3), GlcNAc(?1-?)
Fuc(a1-6), GalNAc(b1-4)
Gal(b1-3), GalNAc(b1-4)
Fuc(a1-3), GalNAc(b1-4)
Gal(b1-3), GalNAc(a1-3)
Gal(b1-4), GalNAc(a1-3)
Fuc(a1-2), GalNAc(a1-3)
Gal(b1-3/4), GalNAc(a1-3)
Gal(?1-3/4/6), Gal(b1-6)
Gal(b1-4), Gal(b1-6)
Gal(a1-3), Gal(b1-6)
6S, Gal(b1-4)
Fuc(a1-6), Gal(b1-4)
Fuc(a1-2), Gal(b1-4)
Fuc(a1-4), Gal(b1-4)
Fuc(a1-2/3/4), Gal(b1-4)
Fuc(a1-3), Gal(b1-4)
Gal(b1-4), Gal(b1-4)
Gal(?1-3/4/6), Gal(b1-4)
Fuc(a1-3/4), Gal(b1-4)
Gal(b1-3), Gal(b1-4)
Gal(a1-3), Gal(b1-4)
Gal(b1-3), Gal(b1-3/4/6)
Gal(b1-3), Gal(b1-3/4)
6S, Gal(b1-3/4)
Fuc(a1-3), Gal(b1-3/4)
Fuc(a1-4), Gal(b1-3/4)
Gal(b1-3/4), Gal(b1-3/4)
Fuc(a1-6), Gal(b1-3/4)
Fuc(a1-2), Gal(b1-3/4)
Gal(?1-?), Gal(b1-3/4)
Fuc(a1-2/3/4), Gal(b1-3/4)
Fuc(a1-3/4), Gal(b1-3/4)
Gal(a1-3), Gal(b1-3/4)
Gal(b1-3), Gal(b1-3)
Fuc(a1-2), Gal(b1-3)
Fuc(a1-4), Gal(b1-3)
Fuc(a1-2/3/4), Gal(b1-3)
Gal(?1-3/4/6), Gal(b1-3)
Fuc(a1-3), Gal(b1-3)
Fuc(a1-3/4), Gal(b1-3)
6S, Gal(b1-3)
Fuc(a1-6), Gal(a1-3)
Fuc(a1-2), Gal(a1-3)
Fuc(a1-2), Gal(?1-?)
Fuc(a1-2), Gal(?1-3/4/6)
Fuc(a1-2), Gal(?1-3)
Fuc(a1-3), Fuc(a1-6)
Fuc(a1-3/4), Fuc(a1-6)
Fuc(a1-6), Fuc(a1-6)
Fuc(a1-2), Fuc(a1-4)
Fuc(a1-3), Fuc(a1-4)
Fuc(a1-2), Fuc(a1-3/4)
Fuc(a1-3), Fuc(a1-3/4)
Fuc(a1-3/4), Fuc(a1-3/4)
6S, Fuc(a1-3/4)
Fuc(a1-2/3/4), Fuc(a1-3/4)
Fuc(a1-2), Fuc(a1-3)
Fuc(a1-3), Fuc(a1-3)
6S, Fuc(a1-2)
−100
−75
−50
−25
0
25
50
75
100
Mean Difference in Abundances of the Intermediates